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In the interest of transparency and to help readers form their own judgements of potential bias, Nature Portfolio journals require authors to declare any competing financial and/or non-financial interest in relation to the work described in the submitted manuscript.

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Large Language Models (LLMs), such as ChatGPT, do not currently satisfy our authorship criteria. Notably an attribution of authorship carries with it accountability for the work, which cannot be effectively applied to LLMs. Use of an LLM should be properly documented in the Methods section (and if a Methods section is not available, in a suitable alternative part) of the manuscript.

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Data availability statement

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- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
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- ☒ We have provided a full data availability statement in the manuscript.

Mandated accession codes ([where applicable](#))

Confirm that all relevant data are deposited into a public repository and that accession codes are provided.

- ☒ All relevant accession codes are provided ☒ Accession codes will be available before publication ☐ No data with mandated deposition

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Code availability statement

For all studies using custom code or mathematical algorithm that is deemed central to the conclusions, the manuscript must include a statement under the heading "Code availability" describing how readers can access the code, including any access restrictions. Code availability statements should be provided as a separate section after the data availability statement but before the References.

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- ☐ ☒ The format shows data distribution clearly (e.g. dot plots, box-and-whisker plots)
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Does not apply

Involved in the study

- ☒ ☐ Macromolecular structural data
- ☒ ☐ Unique biological materials
- ☒ ☐ Research animals and/or animal-derived materials that require ethical approval
- ☒ ☐ Human embryos, gametes and/or stem cells
- ☒ ☐ Human research participants
- ☒ ☐ Clinical data
- ☒ ☐ Archaeological, geological, and palaeontological materials

Macromolecular structural data

Policy information about [special considerations](#) for specific types of data

Validation report

☐ We have provided an official validation report from [wwPDB](#) for all macromolecular structures studied.

Biological materials

Policy information about [availability of materials](#)

Obtaining biological materials

☐ We have described these restrictions in the manuscript. ☐ We have described how to obtain all materials in the manuscript.

Research animals

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research

Ethical compliance

- ☐ We have complied with all relevant ethical regulations and include a statement affirming this in the manuscript.

Ethics committee

- ☐ We have disclosed the name(s) of the board and institution that approved the study protocol in the manuscript.

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- The conditions for donating materials for the research

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Phase 2 and 3 randomized controlled trials

We have provided the [CONSORT checklist](#) with your submission.

- ☐ Yes ☐ No ☐ Not a phase 2/3 randomized controlled trial

Tumor marker prognostic studies

We have followed the [REMARK reporting guidelines](#).

- ☐ Yes ☐ No ☐ Not a tumor marker prognostic study

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I certify that all the above information is complete and correct.

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Jie Yu

Date

4 Apr 2025

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